Worksheet 5C: Graphs of Sine and Cosine — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Basic Curves

1. 0, $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$, 1, 0, -1, 0
2. 1, $\frac{\sqrt{3}}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{1}{2}$, 0, -1, 0, 1
3. $x = \frac{\pi}{4}, \frac{5\pi}{4}$
4. $x = 0, \pi, 2\pi$
5. $x = \frac{\pi}{2}, \frac{3\pi}{2}$
6. Max at $\frac{\pi}{2}$, min at $\frac{3\pi}{2}$
7. Max at $0, 2\pi$, min at π
8. $-\sin(x)$
9. $\cos(x)$
10. (a) $\frac{1}{440}$ s $\approx$ 2.27 ms (b) 440
11. 0
12. 1
13. $t = \frac{1}{1760}$ s

Part B: Amplitude

14. Amplitude = 3
15. Amplitude = 0.5
16. Amplitude = 2, reflected
17. Amplitude = 1, reflected
18. $y = 4\sin(x)$
19. $y = -\frac{1}{3}\sin(x)$
20. (a) 100 times (b) 10,000 times

21. $y = -A\sin(x)$
22. Min: 0.5, Max: 1.5

Part C: Period

23. π
24. $\frac{2\pi}{3}$
25. 4π
26. $\frac{1}{2}$
27. 6
28. $B = 2$
29. $B = \frac{1}{2}$
30. $B = 2\pi$
31. $B = 120\pi$
32. (a) 0.00382 s (b) $y = \sin(1643.5t)$
33. 523.2 Hz
34. (a) 1.02×10^{-8} s (b) 98.5$\times$ shorter period
35. (a) 4.17×10^{-10} s (b) 2×10^{-10} s (c) 5 GHz
36. (a) 2×10^{-15} s (b) 600 nm
37. (a) 10^{-18} s (b) 0.3 nm (c) Wavelength comparable to atomic spacing
38. (a) 1.35×10^{-8} m (b) 4.5×10^{-17} s (c) 2.22×10^{16} Hz
39. (a) 1.40×10^{17} rad/s

Part D: Phase Shift

40. $\frac{\pi}{4}$ right
41. $\frac{\pi}{3}$ left
42. $\frac{\pi}{2}$ right
43. $\frac{\pi}{6}$ left
44. $y = \sin(x - \frac{\pi}{6})$
45. $y = \cos(x + \frac{\pi}{4})$
46. $y = \sin(2(x - \frac{\pi}{4}))$
47. $V_1 = 0$, $V_2 = -\frac{\sqrt{3}}{2}V_0$, $V_3 = \frac{\sqrt{3}}{2}V_0$
48. $\Delta t \approx 1.59 \times 10^{-10}$ s

49. ≈ 4.77 cm

Part E: Vertical Shift

50. A=2, P=2π, C=0, D=3
51. A=3, P=π, C=0, D=-1
52. A=4, P=2π, C=$\frac{\pi}{2}$, D=5
53. A=$\frac{1}{2}$, P=$\frac{\pi}{2}$, C=$-\frac{\pi}{4}$, D=-2
54. (a) 5°C (b) 10°C (c) 365 days (d) $T(t) = -5\cos(\frac{2\pi}{365}(t-14)) + 10$
55. (a) 1.2 m (b) 2.0 m (c) 12.5 hours (d) $h(t) = 1.2\cos(\frac{2\pi}{12.5}t) + 2.0$
56. (a) A=0.6°C, D=36.8°C (c) 36.68°C

Part F: Complete Form

57. A=3, P=π, C=$\frac{\pi}{2}$, D=1
58. A=2, P=4π, C=$-\frac{\pi}{2}$, D=-3
59. $y = 5\sin(\frac{1}{2}(x - \frac{\pi}{2})) - 2$
60. $y = 2\cos(2\pi x) + 3$
61. (a) 440 Hz, 880 Hz, 1320 Hz (b) 1.0, 0.3, 0.1 (c) Harmonics ($2\times$ and $3\times$ fundamental)
62. (a) Odd integers (b) $\frac{1}{n}$ for odd n
63. (a) $2L$ (b) L (c) n
64. (a) 2000 Hz (b) 200 Hz
65. (a) 22,050 Hz (b) Exceeds human hearing range
66. Yes! 70 Hz aliases to $|70 - 100| = 30$ Hz

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